

A Source-Disjoint Audit of QR-Anchor-Gated Residual Transfer for Depth-Two QAOA

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We audit a four-query surrogate for a finite-grid, depth-two quantum approximate optimization algorithm (QAOA) search. The method subtracts the established exact depth-one MaxCut expectation, selects four candidate columns by pivoted QR, and transfers the source residual row nearest to the four measured target residuals. The experiment uses 16 checksum-pinned Stanford GSET graphs: six training, eight development, and two untouched confirmation sources. Temperatures $\tau = 0$ and 2 tie on development to reported precision; their RMSE difference is only 6×10^{-18} and flips under a fresh-wheel replay. We therefore freeze the simpler uniform choice $\tau = 0$ before evaluating G67 and G77. On their 16 confirmation tasks, anchor-gated transfer has surface RMSE 0.006095, smaller than zero-query nearest-depth-one transfer (0.011280), uniform linear interpolation (0.011811), raw-surface interpolation (0.011821), and fixed random anchors (0.035460). All but the random method nevertheless make all 16 grid decisions correctly; the evidence therefore does not show that four queries are necessary for the decision. We prove the established depth-one formula, source separation, linear interpolation and gated-library bounds, finite-grid regret, and explicit work/storage counts. The result is a source-disjoint conditional CPU study, not a new QR/nearest-neighbor method, hardware result, or quantum advantage claim.

I. INTRODUCTION

QAOA alternates a problem Hamiltonian with a mixer and delegates parameter selection to a classical outer loop [1]. Parameter concentration and transfer can reduce that loop for related graph families [2, 3]. A finite grid of N candidate parameter vectors can instead be viewed as a response vector; many source tasks form a response matrix. This makes column skeletons an obvious possible way to query only a few target candidates.

Pivoted-QR point selection is established in Q-DEIM [4]; DEIM-induced CUR gives a related skeleton factorization and error analysis [5]; and low-rank tensor completion has already been proposed for variational-quantum landscapes [6]. Nor is the depth-one MaxCut baseline new: its edge-local expectation is known analytically [7]. These results motivate a stricter question than low-rank surface completion: can a small set of matrix columns identify a reusable source residual more accurately than zero-query transfer and linear column interpolation? The audited method treats anchor values as coordinates for selecting a source row rather than coefficients for a global linear map.

The contribution is an auditable combination and stress test, not a new general matrix primitive. We specify QR-anchor-gated residual transfer, derive conditional error and resource bounds, and compare it against zero-query, linear, raw-surface, random-anchor, and exhaustive rules at equal or lower target-query budgets. Authenticated provenance, complete tables, tests, and four quantitative figures are released. G67 and G77 are read only after freezing the algorithm and numerical-tie rule.

II. AUTHENTICATED FINITE-GRID PROBLEM

For an unweighted simple graph $G = (V, E)$, let

$$C = \sum_{(u,v) \in E} \frac{I - Z_u Z_v}{2}, \quad B = \sum_{u \in V} X_u. \quad (1)$$

At depth two, set

$$U(\boldsymbol{\theta}) = e^{-i\beta_2 B} e^{-i\gamma_2 C} e^{-i\beta_1 B} e^{-i\gamma_1 C}, \quad (2)$$

$$y_G(\boldsymbol{\theta}) = |E|^{-1} \langle + | U(\boldsymbol{\theta})^\dagger C U(\boldsymbol{\theta}) | + \rangle. \quad (3)$$

where $\boldsymbol{\theta} = (\gamma_1, \beta_1, \gamma_2, \beta_2)$. Each coordinate uses four fixed values, so $N = 4^4 = 256$. A matrix-free state-vector kernel applies each one-qubit mixer directly to amplitudes; it never forms a $2^{|V|}$ by $2^{|V|}$ unitary.

The external data are 16 Stanford GSET MaxCut files [8]. Every byte stream is checked against a registered SHA-256 digest. For each source, eight deterministic, seeded breadth-first traversals select ten vertices and form the unweighted induced graph. These 128 objects are *derived tasks*, not independent benchmark instances and not hardware measurements. Table I fixes the protocol.

Proposition 1 (Source separation) *Let*

$$\mathcal{S}_{\text{tr}} = \{G1, G6, G11, G14, G18, G22\}, \quad (4)$$

$$\mathcal{S}_{\text{dev}} = \{G32, G43, G55, G70, G60, G72, G65, G66\}, \quad (5)$$

$$\mathcal{S}_{\text{con}} = \{G67, G77\}. \quad (6)$$

The implemented protocol is source-disjoint: no source byte stream, nor a task derived from it, belongs to more than one group.

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TABLE I. Frozen protocol. Splits are by complete source file; no derived task crosses a split.

Item	Value
Authenticated sources	16 GSET files
Source split	6 train / 8 dev / 2 confirm
Derived tasks per source	8
Task size	10 vertices
Depth-two candidates	$4^4 = 256$
Task split	48 train / 64 dev / 16 confirm
Primary anchor budget	$r = 4$
Response generation	noiseless state vector
Hardware observations	none

Proof. The three displayed sets have pairwise-empty intersections. The task builder assigns a record the split of its sole source name and constructs its vertices only from that source’s authenticated adjacency list. Thus a record cannot cross groups. This establishes data-flow separation, not probabilistic independence among the eight tasks from one source. $\square$

III. EXACT BASELINE AND RESIDUAL ANCHORS

For edge (u, v) define

$$a_{uv} = d_u - 1, \quad b_{uv} = d_v - 1, \quad t_{uv} = |N(u) \cap N(v)|. \quad (7)$$

The following formula is established prior work [7].

Theorem 1 (Exact depth-one formula) *For the convention in Eq. (3), define $c = \cos \gamma$ and $q = \cos(2\gamma)$. Then*

$$F_G^{(1)}(\gamma, \beta) = \sum_{(u,v) \in E} \Phi_{uv}, \quad (8)$$

$$\Phi_{uv} = \frac{1}{2} + \frac{\sin(4\beta) \sin \gamma}{4} (c^{a_{uv}} + c^{b_{uv}}) - \frac{\sin^2(2\beta)}{4} c^{a_{uv}+b_{uv}-2t_{uv}} (1 - q^{t_{uv}}). \quad (9)$$

The normalized baseline is $b_G = F_G^{(1)}/|E|$. After the edge statistics are available, evaluating A distinct (γ, β) pairs costs $O(A|E|)$ arithmetic and $O(|V| + |E|)$ graph storage.

Proof. All cost terms commute. Hence the Heisenberg causal cone of $Z_u Z_v$ contains only cost gates incident on u or v . The two needed conjugations are

$$e^{i\beta X} Z e^{-i\beta X} = Z \cos(2\beta) + Y \sin(2\beta), \quad (10)$$

$$e^{i\theta P/2} Q e^{-i\theta P/2} = Q \cos \theta + iPQ \sin \theta \quad (PQ = -QP). \quad (11)$$

In $|+\rangle^{\otimes |V|}$, only Pauli strings containing I and X survive. The $ZY + YZ$ terms contribute

$$-\frac{\sin(4\beta) \sin \gamma}{2} [(\cos \gamma)^{a_{uv}} + (\cos \gamma)^{b_{uv}}] \quad (12)$$

to $\langle Z_u Z_v \rangle$. Unmatched neighbors in the YY term contribute cosines. Pairing the t_{uv} common neighbors and summing the even subsets by the binomial theorem contributes

$$\frac{\sin^2(2\beta)}{2} (\cos \gamma)^{a_{uv}+b_{uv}-2t_{uv}} [1 - (\cos 2\gamma)^{t_{uv}}]. \quad (13)$$

Substitution into $\langle C_{uv} \rangle = (1 - \langle Z_u Z_v \rangle)/2$ and summation over edges gives Eq. (9). Once (a_{uv}, b_{uv}, t_{uv}) are stored per edge, each angle pair performs constant work per edge; the stated counts follow. $\square$

The implementation compares Eq. (9) with an independent depth-one state-vector path on all generated records. The maximum absolute discrepancy is 8.88×10^{-16} .

Let training depth-two responses and analytic baselines form $Y, B \in \mathbb{R}^{m \times N}$ with $m = 48$, and define the residual matrix

$$R = Y - B. \quad (14)$$

For temperature τ , decision salience is

$$w_j = \frac{N}{m} \sum_{i=1}^m \frac{e^{\tau Y_{ij}}}{\sum_{\ell=1}^N e^{\tau Y_{i\ell}}}, \quad \frac{1}{N} \sum_{j=1}^N w_j = 1. \quad (15)$$

Pivoted QR of $R \text{diag}(\sqrt{w})$ selects r columns J . With $R_J = R(:, J)$, the linear comparator uses

$$X = R_J^\dagger R \in \mathbb{R}^{r \times N}. \quad (16)$$

For a target residual row $f = y - b$, four target evaluations reveal f_J ; linear interpolation returns

$$\hat{y}_{\text{lin}} = b + f_J X. \quad (17)$$

The primary gated rule instead chooses

$$i_*(f) = \min \arg \min_{1 \leq i \leq m} \|f_J - R(i, J)\|_2, \quad (18)$$

$$\hat{y}_{\text{gate}} = b + R(i_*(f), :). \quad (19)$$

The minimum-index convention makes ties deterministic.

The temperature grid $\{0, 2, 4, 8, 12, 20, 40, 80\}$ is resolved only on development sources. Temperatures 0 and 2 have identical regret and RMSE to twelve decimal places; their unrounded RMSE difference, 6×10^{-18} , reverses under a fresh wheel replay. A 10^{-12} numerical-tie tolerance followed by the smallest temperature therefore selects $\tau = 0$ (Fig. 1). Confirmation data are not used in this rule.

Proposition 2 (Residual reconstruction bounds) *Suppose first that R_J has full column rank. Then the matrix in Eq. (16) satisfies $X(:, J) = I_r$. Define*

$$\mathcal{F}_X = \{z \in \mathbb{R}^N : z = z_J X\}. \quad (20)$$

Then $T(f) = f_J X$ interpolates the anchors, is exact for every $f \in \mathcal{F}_X$, and obeys

$$\|f - T(f)\|_\infty \leq (1 + \|X\|_1) \inf_{z \in \mathcal{F}_X} \|f - z\|_\infty. \quad (21)$$

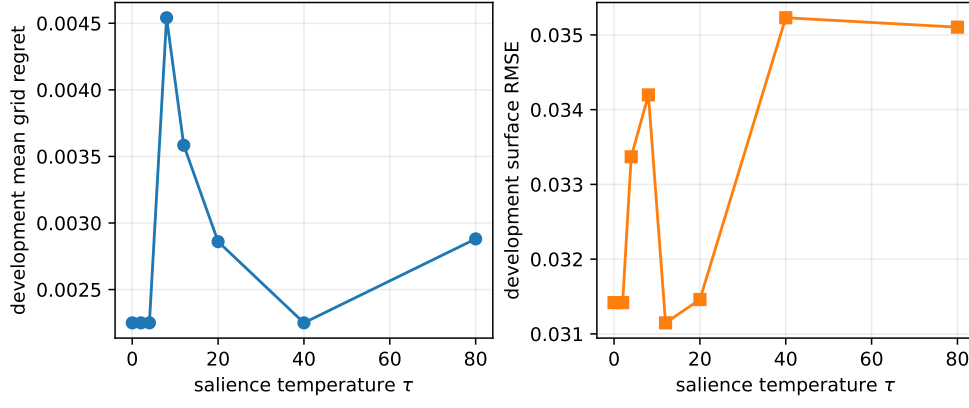


FIG. 1. Development-only temperature audit for the gated rule. Values at $\tau = 0$ and 2 tie within 10^{-12} , so the frozen rule selects the simpler uniform weighting $\tau = 0$. The panels report decision and reconstruction metrics separately.

If additionally $\text{rank}(R) = \text{rank}(R_J) = r$, every row in the training row space belongs to $\mathcal{F}_X$.

For the gated rule, define the target-dependent restriction distortion

$$\kappa_J(f; R) = \max_i \frac{\|f - R(i, :)\|_\infty}{\|f_J - R(i, J)\|_2}, \quad (22)$$

where a zero-over-zero term is defined as zero and a positive numerator over zero is $+\infty$. If $d_\infty(f, R) = \min_i \|f - R(i, :)\|_\infty$, then

$$\|f - R(i_*(f), :)\|_\infty \leq \kappa_J(f; R) \sqrt{r} d_\infty(f, R). \quad (23)$$

Moreover, the gated rule exactly recovers every library row whenever the anchor restriction distinguishes distinct library rows.

Proof. Because $R_J^\dagger R_J = I_r$, $X(:, J) = R_J^\dagger R_J = I_r$, and hence $(T(f))_J = f_J$. If $f \in \mathcal{F}_X$, its definition gives $T(f) = f$. For arbitrary $z \in \mathcal{F}_X$, linearity and the induced matrix norm give

$$\|f - T(f)\|_\infty \leq \|f - z\|_\infty + \|T(f - z)\|_\infty \quad (24)$$

$$\leq \|f - z\|_\infty + \|(f - z)_J\|_\infty \|X\|_1 \quad (25)$$

$$\leq (1 + \|X\|_1) \|f - z\|_\infty. \quad (26)$$

Taking the infimum proves Eq. (21). Under the rank condition, $\text{col}(R_J) = \text{col}(R)$, so $R_J R_J^\dagger R = R$. Therefore $R = R_J X$ and every linear combination of rows of R is fixed by T . For the gated rule, choose $k \in \arg \min_i \|f - R(i, :)\|_\infty$. The definition of i_* and Eq. (22) give

$$\|f - R(i_*, :)\|_\infty \leq \kappa_J \|f_J - R(i_*, J)\|_2 \quad (27)$$

$$\leq \kappa_J \|f_J - R(k, J)\|_2 \quad (28)$$

$$\leq \kappa_J \sqrt{r} \|f - R(k, :)\|_\infty, \quad (29)$$

which is Eq. (23). If f is a library row, its anchor distance is zero; injectivity of the restriction makes every zero-distance choice the same full row. $\square$

These are conditional approximation statements. They neither assert that a new source residual lies near $\mathcal{F}_X$ or the library nor control κ_J before the unqueried target residual is known.

Theorem 2 (Finite-grid decision regret) For a true response $y \in \mathbb{R}^N$, a prediction $\hat{y}$, and $\hat{j} \in \arg \max_j \hat{y}_j$, the regret satisfies

$$0 \leq \max_j y_j - y_{\hat{j}} \leq 2\|y - \hat{y}\|_\infty. \quad (30)$$

In particular, if the true best candidate has margin $\Delta = y_{j_*} - \max_{j \neq j_*} y_j$ and $\|y - \hat{y}\|_\infty < \Delta/2$, the predicted maximizer is j_* .

Proof. Since $\hat{y}_{\hat{j}} \geq \hat{y}_{j_*}$,

$$y_{j_*} - y_{\hat{j}} = (y_{j_*} - \hat{y}_{j_*}) + (\hat{y}_{j_*} - \hat{y}_{\hat{j}}) + (\hat{y}_{\hat{j}} - y_{\hat{j}}) \quad (31)$$

$$\leq |y_{j_*} - \hat{y}_{j_*}| + |\hat{y}_{\hat{j}} - y_{\hat{j}}| \quad (32)$$

$$\leq 2\|y - \hat{y}\|_\infty. \quad (33)$$

If the infinity error is below $\Delta/2$, then for every $j \neq j_*$, $\hat{y}_{j_*} - \hat{y}_j > \Delta - 2\|y - \hat{y}\|_\infty > 0$. $\square$

Proposition 3 (Explicit resource contract) For $m \leq N$ training tasks and $r \leq m$ anchors, standard dense QR with column pivoting selects the gated anchors in

$$O(Nm^2) \text{ work}, \quad O(mN) \text{ storage}, \quad (34)$$

in addition to the mN training response evaluations. For one target, anchor gating uses exactly r depth-two evaluations, $O(mr + N)$ transfer work, and $O(mN + N)$ model/workspace storage, plus the analytic depth-one work in Theorem 1. Constructing the optional linear map adds $O(mrN + mr^2)$ work and $O(rN)$ storage; applying it costs $O(rN)$. Exhaustive search uses N depth-two evaluations. A nearest-depth-one transfer uses zero target depth-two evaluations but $O(mN)$ comparison work and $O(mN)$ stored training data.

TABLE II. Work and storage excluding the cost of one depth-two response. $A \leq N$ is the number of distinct first-layer pairs.

Method	$p = 2$	Work	Model
Depth-one only	0	$O(A E + N)$	$O(N)$
Mean residual	0	$O(A E + N)$	$O(N)$
Nearest $p = 1$	0	$O(A E + mN)$	$O(mN)$
Anchor-linear	r	$O(A E + rN)$	$O(rN)$
Anchor-gated	r	$O(A E + mr + N)$	$O(mN)$
Exhaustive	N	$O(N)$ bookkeeping	$O(N)$

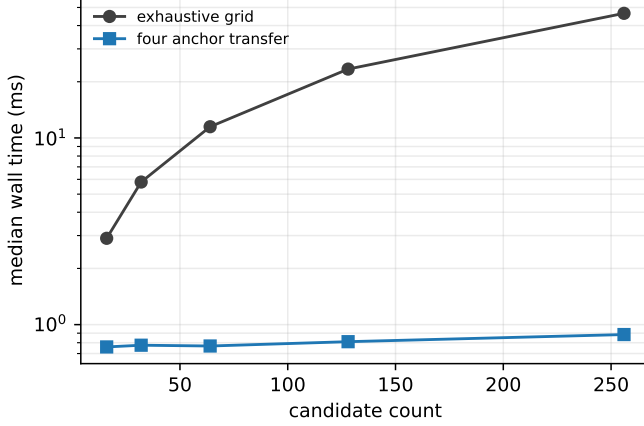


FIG. 2. Single-host implementation timing on one ten-vertex confirmation task. The vertical axis is logarithmic. These medians illustrate the query contract; the asymptotic counts are those of Proposition 3.

Proof. QRCP of an m by N matrix with $m \leq N$ costs $O(Nm^2)$. Gating compares m restricted rows of length r , then copies one length- N row, giving $O(mr + N)$. The library contains mN values. Solving the optional m by r least-squares problem with N right-hand sides costs $O(mr^2 + mrN)$ and stores rN values. Obtaining f_J requires exactly the r selected depth-two responses because its baseline entries are analytic. The exhaustive and zero-query counts follow by direct enumeration. $\square$

Table II makes these offline and target-time resource counts comparable under one convention.

IV. RESULTS: THE STRONGER BASELINE CHANGES THE CONCLUSION

We report normalized surface RMSE, grid regret, and exact decision fraction. The comparators are fixed before reading confirmation results: (i) depth-one only; (ii) the mean training residual; (iii) the residual of the training task nearest in analytic depth-one-surface RMSE; (iv) uniform-QR linear reconstruction; (v) QR-anchor-gated residual transfer; (vi) fixed random anchors; and (vii) uniform-QR reconstruction of the raw surface.

On development, anchor gating improves the zero-query nearest rule’s RMSE by 0.008045 and mean regret by 0.000626, while adding one exact decision out

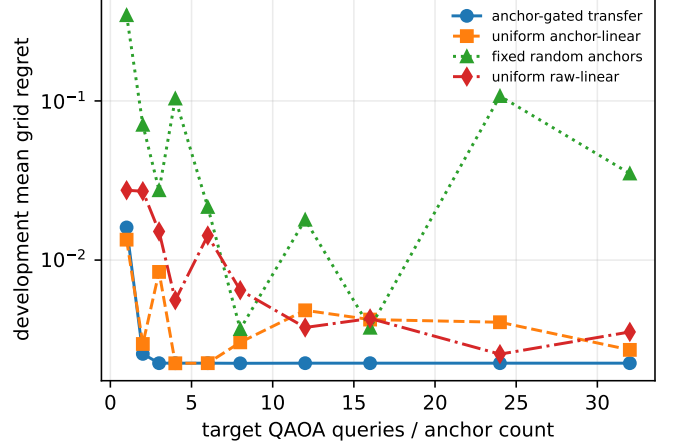


FIG. 3. Development mean regret versus target query count. The registered primary budget is four. Nonmonotonic curves preclude a generic “more rank is better” claim.

of 64. Uniform linear interpolation has regret lower by only 4.1×10^{-6} but RMSE higher by 0.001750. Figure 3 shows that the ordering is rank-dependent and nonmonotone; more anchors do not uniformly improve regret.

On confirmation, anchor gating has the smallest nonexhaustive surface RMSE (Fig. 4). Yet the mean-residual and nearest-depth-one rules use zero depth-two queries and choose the same exact grid maximizer on all 16 tasks; uniform linear and raw-linear rules also score 16/16. Thus the confirmation evidence supports better surface reconstruction from the four anchors, but it cannot attribute correct decisions to those queries. Because the confirmation set contains only G67 and G77, 16/16 is descriptive rather than a population guarantee.

V. MEASURED RUNTIME AND ITS LIMITS

Figure 2 times the actual Python kernels on one authenticated G67-derived ten-vertex task under macOS arm64 and Python 3.13.12. Each point is the median of seven repetitions after one warm-up. At $N = 256$, exhaustive state-vector evaluation takes 46.512 ms, while four state-vector evaluations, analytic baseline construction, and the gated transfer take 0.885 ms. On this particular task the transferred surface agrees pointwise with exhaustive evaluation to 2.22×10^{-16} . This isolated timing is not an equal-accuracy solver speedup claim across confirmation tasks. It is also not a hardware claim: shot noise, compilation, batching, and communication are absent.

The state vector costs $O(2^{|V|})$ memory and $O(p|V|2^{|V|} + p|E|2^{|V|})$ arithmetic per response in this implementation. Nothing here improves that per-query complexity. Equation (19) only conditionally substitutes r such responses for N responses after an offline matrix has been learned.

TABLE III. Source-disjoint results. “Queries” counts target depth-two responses. Development selected τ ; confirmation did not select any method or hyperparameter. Lower regret/RMSE and higher exact fraction are better; the confirmation comparison discussed below is read from Table III.

Method	Queries	Development: 64 tasks			Confirmation: 16 tasks		
		Regret	RMSE	Exact	Regret	RMSE	Exact
Depth-one only	0	.068455	.092466	.000	.057170	.071402	.000
Mean residual	0	.008026	.060075	.625	.000000	.049454	1.000
Nearest $p = 1$ residual	0	.002876	.039463	.828	.000000	.011280	1.000
Uniform anchor-linear	4	.002246	.033168	.844	.000000	.011811	1.000
Anchor-gated residual	4	.002250	.031418	.844	.000000	.006095	1.000
Fixed random anchors	4	.103504	.117963	.344	.019824	.035460	.625
Uniform raw-linear	4	.005597	.038563	.750	.000000	.011821	1.000

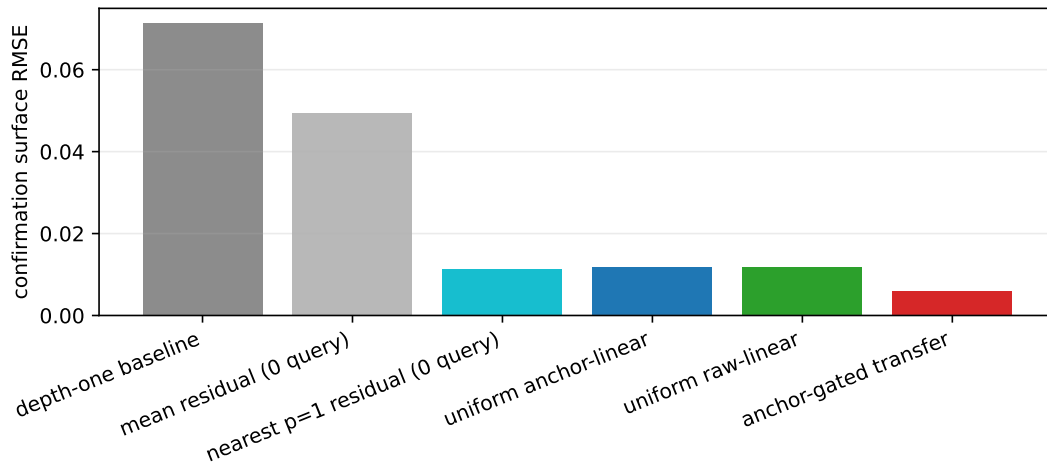


FIG. 4. Confirmation surface RMSE. QR-anchor gating is best among the nonexhaustive comparators. It and the zero-query nearest- $p = 1$ rule nevertheless make the same 16/16 grid decisions, separating reconstruction accuracy from evidence of decision-query necessity.

VI. REPRODUCIBILITY AND CLAIM BOUNDARY

The public package authenticates all 16 source files, recomputes the 128 state-vector landscapes, verifies the locked JSON without overwriting it, regenerates four figures, and tests the formulas and bounds. The deterministic entry point is

`qaoa-anchor-reproduce -verify.`

Timing is intentionally separate because wall time is host-specific: `qaoa-anchor-benchmark`. The manifest records hashes and byte counts for scientific tables, the locked result, timing evidence, and every PDF/PNG figure.

The following are outside the evidence: continuous-angle optimization; graph sizes above ten qubits; weighted MaxCut; noise or quantum hardware; MaxCut approximation ratios or global optima; statistical independence of tasks from one source; and generalization be-

yond the two confirmation sources. The numerical tests cannot prove the analytic results; they check implementations and exhibited consequences. The proofs above carry the mathematical claims.

VII. CONCLUSION

QR-anchor gating is a coherent conditional surrogate: it reduces target depth-two evaluations from N to r , improves development RMSE and regret over zero-query transfer, and has the smallest confirmation surface RMSE. But the source-disjoint confirmation data do not show that its four queries are necessary for the finite-grid decision: two zero-query residual transfers make every confirmation decision correctly. The defensible outcome is a conditional reconstruction result and a sharper next experiment: add many untouched source families, larger classically tractable graphs, noise-aware measurements, and predeclared decision baselines before asking whether anchor queries add out-of-source decision value.

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